Julian Lüken · Am Hachweg 1 · 37083 Göttingen · Germany

University of Antwerp
Prinsstraat 13
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PhD student – 3D tomography of nano-samples

Dear recruitment team,

I am writing in response to your advertisement for the PhD vacancy in 3D tomography of nano-samples. I believe my experience, skills and qualification are not only an asset, but an ideal match to the requirements of your operation.

I currently live in Göttingen, where I very recently finished my M.Sc. studies in Applied Computer Science at the Georg-August University. My topic of specialization is Scientific Computing. During my studies I took three consecutive courses on Mathematical Signal and Image Processing at the Institute of Numerical and Applied Mathematics, the last one of which taught computer tomography. I am therefore not only well-acquainted with Fourier analysis, but also with the Radon transform, which is the historical foundation of modern computer tomography.

Presently, I work at the Institute of Propulsion Technology at the German Aerospace Center, where I wrote my master's thesis in cooperation with my university. In the course of this thesis project about programming a turbine blade cooling geometry generator called CoolingGen, I could prove my deep mathematical understanding of linear algebra and my implementation talent in C++ and Python. To assess the quality of certain features of the software, CoolingGen's creation also required me to communicate with aerospace engineers, which was always a productive yet pleasant experience.

Even though I whole-heartedly enjoy my current work, I feel that I have to move on. Computer tomography is an amalgamation of various interesting sciences, a life-saving medical instrument and an amazing assessment tool in my previous line of work. Given that this position is also in the beautiful city of Antwerp, I must apply for it.

Kind regards,

Lüken

Julian Lüken

Julian **Lüken**

Expert in Signal Processing and Applied Geometry

Education

Qualification

2019–2023 | Applied Computer Science (M. Sc.)
Specialisation in Scientific Computing and Signal Processing
Georg-August University Göttingen

2016–2019 | Applied Computer Science (B. Sc.)
Specialisation in Scientific Computing
Georg-August University Göttingen

2014–2016 | Mathematics (B.Sc.)
Discontinued
Georg-August University Göttingen

2013 | Abitur
Europaschule Gymnasium Rhauderfehn

Academic Projects

2022 | Master thesis project **CoolingGen**
A parametric CAD software using NURBS, specifically designed for generating turbine blade cooling geometries
Graded 1.0
German Aerospace Center, Göttingen

2019 | Bachelor thesis project **SUSAN**
Image feature detection based on a combinatorial scheme
Graded 1.1
Institute of Numerical and Applied Math, Göttingen

2018 | Scientific Computing Practical
Denoising of piecewise constant images
Graded 1.0
Institute of Numerical and Applied Math, Göttingen

Practical Experience

Career

2021–2023 | Research Associate, Developer
German Aerospace Center, Göttingen

2019–2021 | IT Department Staff Member
keydel bock ingenieure gmbh, Göttingen

Jobs

2017–2019 | Postman
Deutsche Post AG, Göttingen

2008–2012 | IT Temp
Gebr. Bruns Heizkessel GmbH, Saterland

Skills

Programming Languages	Long-time experience with Matlab, Python, C, C++
Digital Signal Processing	Processing tools (opencv, scipy) and visualization tools (matplotlib)
Computer-Aided Design	Programming with OpenCASCADE or my personal C++ CAD kernel (visualization with FreeCAD)
Math	Excellent knowledge of Geometry, Linear Algebra and Fourier Analysis
Languages	German (native), English (CEFR C1)
Misc	LaTeX, Javascript, HTML, CSS, git, bash

Interests

Hiking, playing the guitar, cooking, reading, going out, playing video games.

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